MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 726C

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex oscillator, integral cavity, tunable frequency 2, 700 to 2, 960 MHz, CW operation

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Ik	F	Alt
Unit:	V	Vdc	Vdc	mAdc	MHz	ft
Maximum:	6.8	330	-250	35	2, 960	10, 000
Minimum:	5.8	---	-30	---	2, 700	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Coated unipotential
Mounting position:	Any

TEST CONDITIONS: See note 1

Parameter:	Ef	Ers	Er	F
Unit:	V	Vdc	Vdc	MHz
	6.3	300	-45 to -135	2, 800 ± 0.3%

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection					
1031	High-frequency vibration	F = 50; 10G; t = 120; no voltages	---	---	---	---
4280	Electronic tuning range	F = 2,800 MHz ±0.3%; Er/50% max Po	ΔF	25	---	MHz
4027	Temperature coefficient	F = 2,800 MHz ±0.3% (see note 2)	Cocf	-0.10	+0.05	MHz/°C
	Quality conformance inspection, part 1 (see note 4)					
1256	Electrode current (cathode)		Ik	---	30.0	mAdc
4214	Cathode emission	Ef = 5.8V; t = 2 minutes	ΔIk/Ik	---	15.0	%
4213	Reflector voltage	Er/max Po; F = 2,800 MHz ±0.3%	Er	-75	-135	Vdc
---	Tunable frequency	Er = -90 to -165 Vdc; high frequency	F	2,960	---	MHz
		Er = -45 to -120 Vdc; low frequency	F	---	2,700	MHz
(C) 4250	Power output	Ef = 6.0 V; Er/max Po;	---	---	---	---
		F = 2,700 MHz	Po	30	---	mW
		F = 2,800 MHz	Po	85	---	mW
		F = 2,960 MHz	Po	50	---	mW
	Quality conformance inspection, part 2					
1211	Insulation of electrodes	300 Vdc; tube cold	Rh-rs	2	---	Meg
			Rk-rs	2	---	Meg
4229	Total reflector current	See note 3	Ir	---	7.0	μAdc
4229	Reflector-leakage current	See note 3	Ir	---	5.0	μAdc
4229	Reflector-gas current	See note 3	Ir	---	2.0	μAdc
1301	Heater current	Ef = 6.3 V	If	410	470	mA
1336	Heater-cathode leakage	Ehk = ±45 Vdc	Ihk	0	100	μAdc
(C) 4280	Electronic tuning range	F = 2,800 MHz ± 0.3%; Er/50% max Po	ΔF	25	---	MHz
4231	Electronic tuning hysteresis (1)	Er/max Po	Ratio	---	0.25	---
	Quality conformance inspection, part 3					
---	Life-test provisions	Group C; power output; F = 2,800 MHz	t	500	---	hrs
---	Life-test end points					
4250	Power output	Er/max Po; Ef = 6.0 V; F = 2,800 MHz	Po	50	---	mW

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NOTES:

1. The tube shall be mounted in a suitable socket and shall be affixed firmly by means of clamps which make contact with the tube only at the flange adjacent to the base. The measurements on the tubes in an oscillating state shall be made using equipment as specified on figure 2, or equal.

2. For the measurement of temperature coefficient, the tube shall be adjusted in free air to the frequency indicated under test conditions. The reflector voltage shall be adjusted for maximum power for the mode specified. If necessary, the mechanical tuning may be cycled to relieve strains which would otherwise release themselves during the temperature cycle. These strains may be released by flexing the tuner bows a number of times. The tube shall be mounted to have that portion above the socket completely inclosed with a metallic cover with an approximate clearance of 1/2 inch (12.7 mm) between the tube and the cover. The walls of the container shall be heated so that an ambient temperature of 80° to 90°C is obtained within. The frequency of oscillation is determined and the walls then cooled at a rate such that the fall of ambient temperature within does not exceed 2°C per minute. When the ambient temperature has been reduced to 25°C or lower, the frequency shall then be recorded. This procedure shall be repeated three times with no readjustment of the frequency on the second two runs. The temperature coefficient shall be computed in frequency units per degree centigrade from the average obtained on the last two runs.

3. This test to be performed at the conclusion of the holding period.

4. The AQL for all tests performed under quality conformance inspection, part 1, shall be 1.0 percent with an inspection level of II.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MU
Navy -
Air Force - 11, 70, 85
DSA - ES

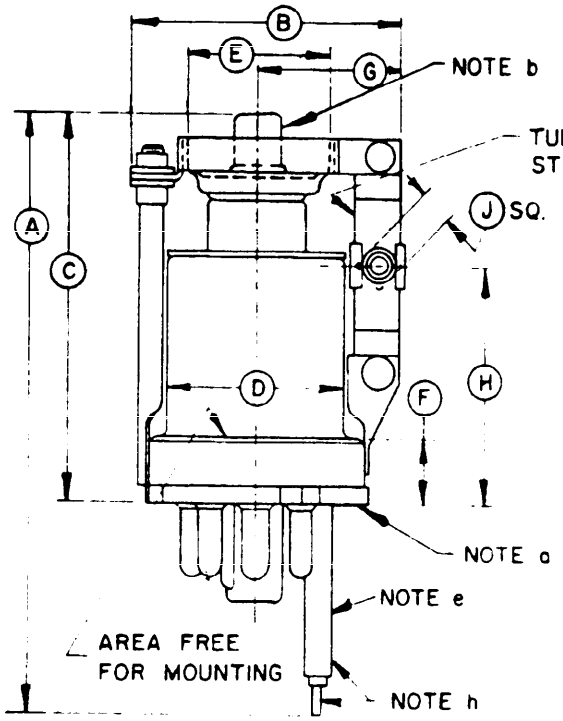
User activities:
Army - AV, ME, WC
Navy - AS, OS, MC, CG, SH
Air Force -

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2392)

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PIN CONNECTIONS	
PIN NO.	ELEMENT
1	sh
2	h
4	OUTPUT LEAD
7	h
8	GUN
CAP	r

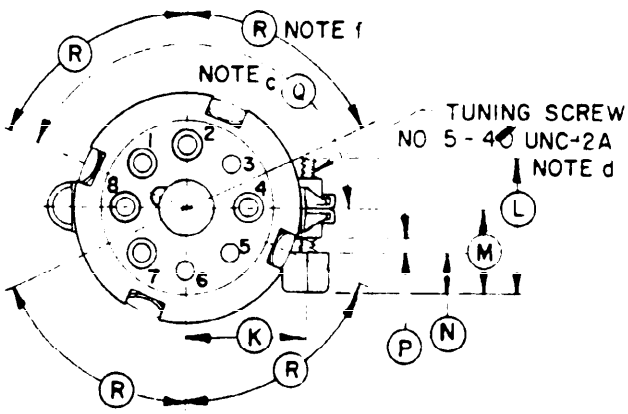


FIGURE 1. Outline drawing of electron tube type 726C.

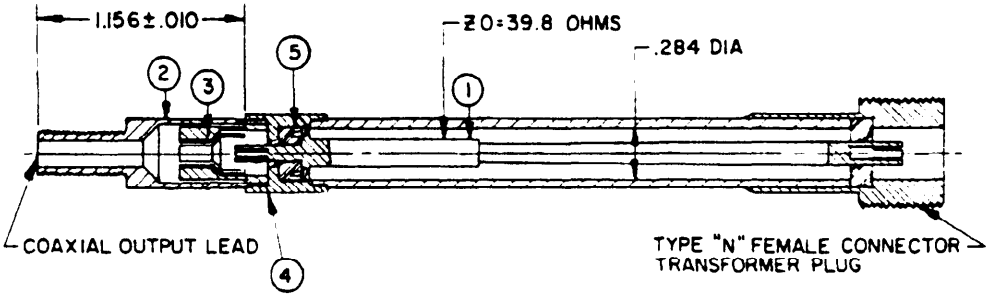
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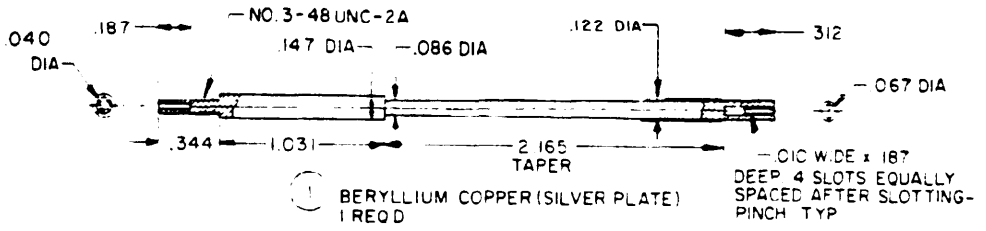
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
F	.330 (8.38)	.400 (10.16)
H	1.313 (33.35)	1.437 (36.50)
K	.656 (16.66)	.718 (18.24)
Q	154° 30'	160° 30'
Quality conformance inspection, part 2		
A	3.188 (80.98)	3.625 (92.08)
B		1.609 (40.87)
C	2.000 (50.80)	2.375 (60.33)
D	1.000 (25.40)	1.015 (25.78)
G		.859 (21.82)
J	.182 (4.62)	.192 (4.88)
M	.485 (12.32)	.516 (13.11)
N	.203 (5.16)	.234 (5.94)
P		.116 (2.95)
Quality conformance inspection, part 3 (periodic check)		
E	.767 (19.48)	.797 (20.24)
L	.766 (19.46)	.797 (20.24)
R	70°	

NOTES:
a. Base: B6-90.
b. Cap: C1-4.
c. Applies to centerline of coaxial lead and centerline of tuner through struts.
d. Shall be lubricated with oildag, or equivalent, non-corrosive lubricant and shall be capable of operation throughout entire range without binding.
e. Nickel (30 MSI silver plate permissible).
f. Area free for mounting.
g. For pin alinement use gage GB8-2.
h. Shall be concentric within .010 (.25 mm).

FIGURE 1. Outline drawing of electron tube type 726C - Continued.

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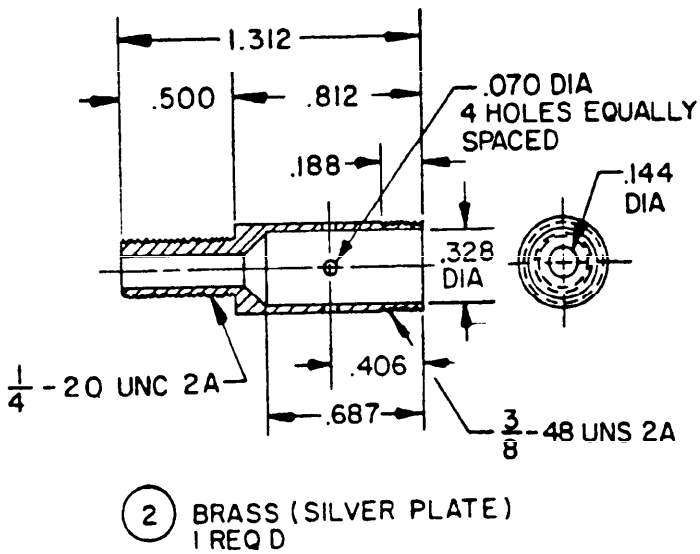




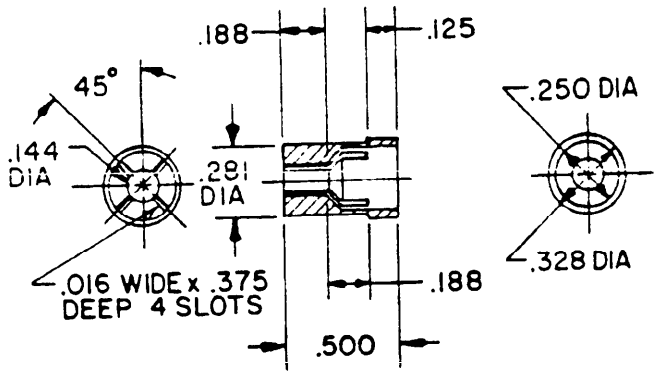
INCHES	MM
.010	.25
.040	1.02
.067	1.70
.086	2.18
.147	3.73
.187	4.75
.284	7.21
.312	7.92
.344	8.74
1.031	26.19
1.156	29.36
2.165	54.99

FIGURE 2. Transducer.

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(2) BRASS (SILVER PLATE)
1 REQ D

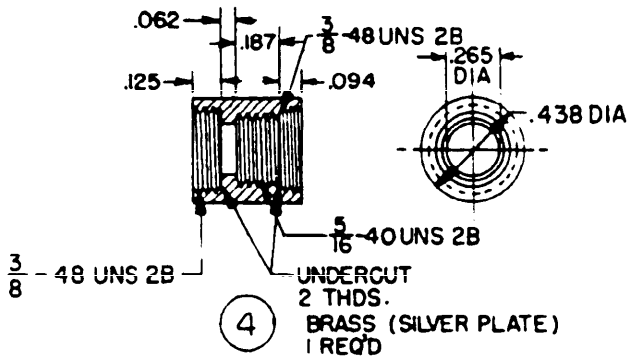


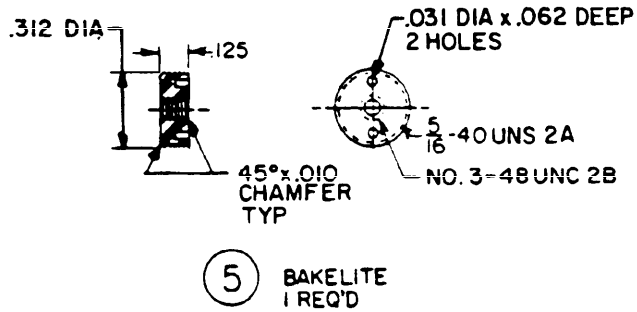
(3) BRASS (SILVER PLATE)
1 REQ D

INCHES	MM
.016	.41
.070	1.78
.125	3.18
.144	3.66
.188	4.78
.250	6.35
.281	7.14
.328	8.33
.375	9.53
.406	10.31
.500	12.70
.687	17.45
.812	20.62
1.312	33.32

FIGURE 2. Transducer - Continued.

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INCHES	MM
.010	.25
.031	.79
.062	1.57
.094	2.39
.125	3.18
.187	4.75
.265	6.73
.312	7.92

NOTES:
a. Dimensions are in inches.
b. Unless otherwise specified tolerance is ±.005 (.13 mm).

FIGURE 2. Transducer - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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